48730-32548, Cyber Security Week-9

Iptables – The Linux Firewall & JUNOS: Firewall User Authentication

Challenge:

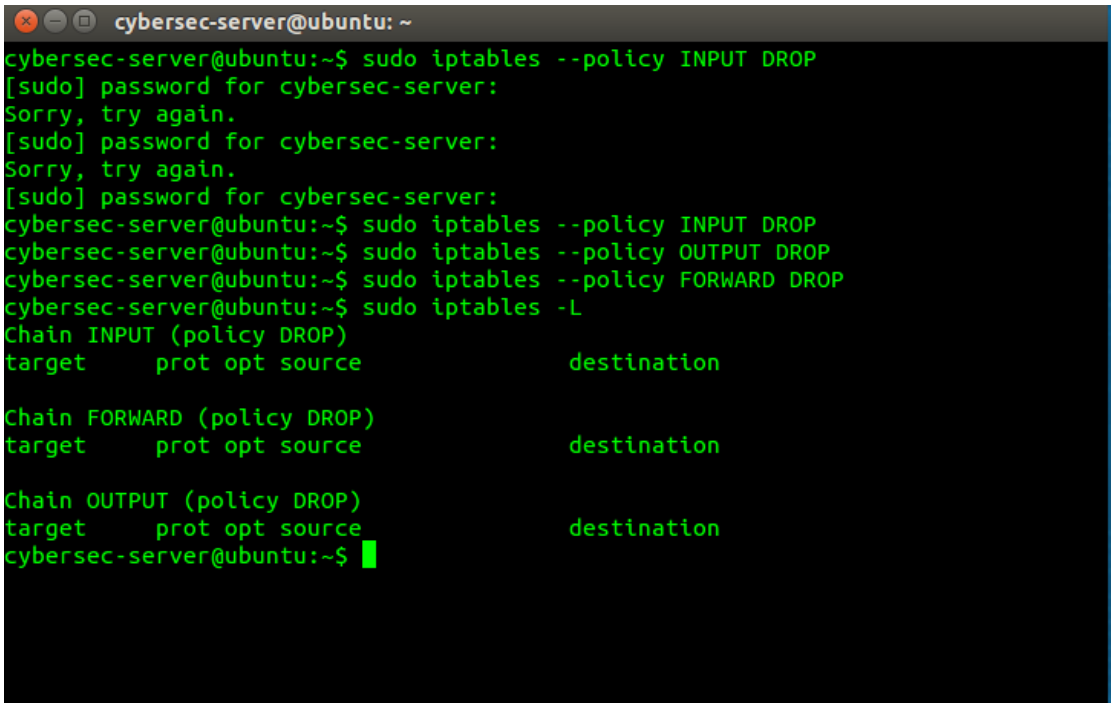
Configure iptables on **Cybersec-server** to **permit HTTP and HTTPS request, drop all other traffic.**

Hints:

1. HTTP and HTTPS use TCP.
2. The rules do not require mention of source or destination address.
3. Connection state: NEW, ESTABLISHED
4. Port Number of HTTP: 80
5. Port Number of HTTPS: 443

Screenshot Required:

Commands used to configure Iptables on Cybersec-Server to permit HTTP, HTTPS and drop other traffic.



```
cybersec-server@ubuntu: ~
cybersec-server@ubuntu:~$ sudo iptables --policy INPUT DROP
[sudo] password for cybersec-server:
Sorry, try again.
[sudo] password for cybersec-server:
Sorry, try again.
[sudo] password for cybersec-server:
cybersec-server@ubuntu:~$ sudo iptables --policy INPUT DROP
cybersec-server@ubuntu:~$ sudo iptables --policy OUTPUT DROP
cybersec-server@ubuntu:~$ sudo iptables --policy FORWARD DROP
cybersec-server@ubuntu:~$ sudo iptables -L
Chain INPUT (policy DROP)
target     prot opt source               destination

Chain FORWARD (policy DROP)
target     prot opt source               destination

Chain OUTPUT (policy DROP)
target     prot opt source               destination
cybersec-server@ubuntu:~$
```

Iptables chain after adding the rules (sudo iptables -L)

```
num  target     prot opt source               destination
cybersec-client@ubuntu:~$ sudo iptables -A INPUT  -p tcp --dport 80  -m state --state NEW,ESTABLISH
ED -j ACCEPT
cybersec-client@ubuntu:~$ sudo iptables -A OUTPUT -p tcp --sport 80  -m state --state ESTABLISHED
       -j ACCEPT
cybersec-client@ubuntu:~$ sudo iptables -A INPUT  -p tcp --dport 443 -m state --state NEW,ESTABLISH
ED -j ACCEPT
cybersec-client@ubuntu:~$ sudo iptables -A OUTPUT  -p tcp --sport 443 -m state --state NEW,ESTABLIS
HED -j ACCEPT
cybersec-client@ubuntu:~$ sudo iptables -L
Chain INPUT (policy DROP)
target     prot opt source               destination
ACCEPT     tcp  --  anywhere             anywhere             tcp dpt:http state NEW,ESTABLISHED
ACCEPT     tcp  --  anywhere             anywhere             tcp dpt:https state NEW,ESTABLISHED

Chain FORWARD (policy DROP)
target     prot opt source               destination

Chain OUTPUT (policy DROP)
target     prot opt source               destination
ACCEPT     tcp  --  anywhere             anywhere             tcp spt:http state ESTABLISHED
ACCEPT     tcp  --  anywhere             anywhere             tcp spt:https state NEW,ESTABLISHED
cybersec-client@ubuntu:~$
```

Trash

JUNOS: Firewall User Authentication

Create Access Profile

1. Enter edit access (Enters the access configuration hierarchy)
2. Enter edit profile <name> (Creates or modifies an access profile with the specified name. Access profiles define authentication settings for users)
3. Enter edit client <name> (Specifies the client name for the access profile. This is used to identify the user or group of users.)
4. Enter edit firewall-user (Configures the firewall user settings within the access profile.)
5. Enter set password <password> (Sets the password for the firewall user. This is the credential users will use for authentication.)
6. enter top (Returns to the top-level configuration hierarchy)
7. Enter show access and look for the policy you created (screenshot)

```
profile Saif {
    client Saif {
        firewall-user {
            password "$9$lxEKLx-VwgaZdVP5z39CLx7NYgik."; ## SECRET-DATA
        }
    }
}
```

```
user@srx# show access
profile profile-name {
    client client-name {
        firewall-user {
            password encrypted-password; ## SECRET-DATA
        }
    }
}
```

Associate Access Profile with pass-through authentication and add banner

1. Enter edit access
2. Enter edit firewall-authentication (Configures firewall authentication settings)
3. Enter set pass-through default-profile <name> (Associates the default access profile with pass-through authentication. Pass-through authentication prompts users for credentials when accessing resources in another zone.)
4. Enter set pass-through telnet banner success “Login Successful” (Configures a success message ("Login Successful") to be displayed after successful authentication via Telnet.)
5. Enter show (screenshot)

```
student@ROUTER_24# show
pass-through {
    default-profile Saif;
    telnet {
        banner {
            success "Login successful!";
        }
    }
}
```

```
user@srx# show
pass-through {
    default-profile profile-name;
    telnet {
        banner {
            success "Login successful!";
        }
```

Configure policy action with Firewall authentication

1. Enter top
2. Enter edit security policies (Enters the security policies configuration hierarchy.)
3. Enter edit from-zone Internet to-zone Internal (Specifies the source and destination zones for the policy.)
4. Enter edit policy <name> (Creates or modifies a policy with the specified name.)
5. Enter set match source-address <IP address/any>
6. Enter set match destination-address <IP address/any>
7. Enter set match application <junos-application> (In this task, use junos-telnet)
8. Enter set then permit firewall-authentication pass-through client-match <name>
9. Enter up (Moves up one level in the configuration hierarchy) 10. Enter show (screenshot)

```
policy Saif {
    match {
        source-address any;
        destination-address any;
        application junos-telnet;
    }
    then {
        permit {
            firewall-authentication {
                pass-through {
                    client-match Saif;
                }
            }
        }
    }
}
```

```
user@srx# show
policy policy-name {
        match {
            source-address <IP address>;
            destination-address <IP address>;;
            application junos-telnet;
        }
        then {
            permit {
                firewall-authentication {
                    pass-through {
                        client-match client-name;
                    }
                } …
```

These commands collectively set up user authentication, associate it with passthrough authentication, and enforce it through security policies. This ensures that only authenticated users can access specific resources, enhancing network security.

Question 1: Please provide screenshots of the access profile, firewall authentication and security policy that you created.

```
profile Saif {
    client Saif {
        firewall-user {
            password "$9$lxEKLx-VwgaZdVP5z39CLx7NYgik."; ## SECRET-DATA
        }
    }
}
```

```
student@ROUTER_24# show
pass-through {
    default-profile Saif;
    telnet {
        banner {
            success "Login successful!";
        }
    }
}
```

```
policy Saif {
    match {
        source-address any;
        destination-address any;
        application junos-telnet;
    }
    then {
        permit {
            firewall-authentication {
                pass-through {
                    client-match Saif;
                }
            }
        }
    }
}
```

Question 2: What are the key benefits of firewall authentication in protecting sensitive information and maintaining network security?

By limiting access to network resources to authorized users, firewall authentication shields private data from unwanted access. It promotes compliance through audit logs, gives administrators the opportunity to implement fine-grained security policies for improved control and protection, and establishes accountability by connecting actions to user identities.